Research Statement

Anand Bhattad, PhD Student, Computer Science, UIUC

The intersection of computer vision, computer graphics, computational photography, and machine learning is at the heart of my research, with a focus on developing self-supervised artificial intelligence systems that can independently and autonomously comprehend the visual world and create a compositional structure for easy image understanding, editing, and manipulation. While current techniques rely heavily on large amounts of labeled data, self-supervised learning makes use of the vast potential of unlabeled data and has demonstrated impressive results in recognition tasks. However, current approaches still face challenges in scenes under varying environmental conditions, such as lighting, weather, location, and time of day [7, 21, 14].

A key limitation in the advancement of current AI systems is the lack of datasets under diverse environmental conditions, which are essential for building systems suitable for critical applications such as self-driving cars and infrastructure or climate change assessments. Additionally, addressing this issue offers opportunities for the creation of new types of content and visual-editing applications. To address these challenges, my research aims to create learning systems that can produce rich, diverse sets of images that are difficult to collect and beyond the capabilities of current AI systems. Specifically, my current research focuses on two main themes: (1) incorporating illumination physics into existing models to expand their capacity and (2) improving the robustness of current methods to allow for better generalization to unseen environments. These two themes are complementary, as both are necessary for creating systems that can handle a wide range of environmental conditions and extend the capabilities of current AI systems. In the following sections, I will elaborate on my key contributions to these themes and discuss future directions.

1 Enhancing AI Capabilities through Illumination Physics

To incorporate illumination physics into existing models, I have developed a novel image decomposition technique that is essential for preserving scene properties that are otherwise difficult to maintain using traditional methods that decompose images into albedo and shading [15]. These traditional methods often lead to unrealistic results, such as the loss of surface details or changes in the spatial layout of the scene. To address these challenges, my technique modifies the traditional albedo-shading decomposition by separating persistent maps analogous to albedo, which encode materials and sharp edges, from transient maps analogous to shading, which encode diffuse and specular shading without geometry (surface normals) information.

This technique is based on the Retinex theory [22] and involves the use of Mondrian with fine edges to encode persistent maps and smooth Perlin noise containing white blobs to encode transient maps. The use of these specific patterns allows for the separation of image information into light-dependent and light-independent components, enabling the realistic manipulation of lighting in images and scenes without altering their albedo or geometry. By imposing decomposition consistency on image parts, this technique allows for the successful transfer of objects from one image to another without losing any surface details and enables the realistic relighting of complex indoor and outdoor scenes. I have applied my novel image decomposition technique to develop two state-of-the-art (SOTA) object insertion methods: DIPR [6] and Illuminator [2]. Both methods are capable of abstracting scene lighting and decoding images with realistic novel object insertion into scenes, allowing for the successful transfer of objects from one image to another without losing any surface details.

In addition to object insertion, I have also developed the first single image relighting method, StyLitGAN [3], which can generate realistic scenes with arbitrary lighting, including cast shadows,



StyleGAN Relighting - 1 Relighting - 2 Relighting - 3 Relighting - 4 Relighting 5

Figure 1: Left: images generated by a vanilla StyleGAN [19]. Others are relightings of the scenes in the leftmost image from our StyLitGAN. Note extreme changes in the illumination of the scene; the geometry and albedo of the scenes do not change; and the images are realistic

inter-reflections, and soft shadows without requiring any physical models or other CGI (computer-generated imagery) labeled data (Figure 1). This is a significant improvement over previous methods, which often require extensive physical modeling or rely on large amounts of labeled data to produce similar results [8]. StyLitGAN can achieve these results by preserving scene semantics, ensuring that the resulting images are not only photorealistic but also semantically accurate. As well as this, it can add or remove light sources and provide a simple alternative to more sophisticated technologies that are currently available.

My research on enhancing AI capabilities with illumination physics has important implications for a wide range of applications, including CGI in film and video games, augmented reality experiences, and virtual reality experiences. My image decomposition technique can be used to improve the realism of these applications, enabling the creation of more immersive and believable virtual worlds. This technique can also be used to create new types of content, such as virtual advertisements that can be inserted into live video streams or video games with dynamic lighting. These methods can help to create more realistic and convincing virtual environments for users to interact with, improving the overall user experience.

2 Improving Robustness and Generalization Capabilities

In my research on improving the robustness and generalization of AI systems, I have developed several techniques and approaches that address critical challenges in this area. One major challenge in the field is the lack of robustness to changes in lighting conditions. Current SOTA dense prediction models are sensitive to lighting changes, resulting in inaccurate predictions. Additionally, these models also fail to capture equivariant properties, meaning they are not able to accurately predict depth and surface normals when the input image is transformed, such as when the image is cropped (Figure 2).

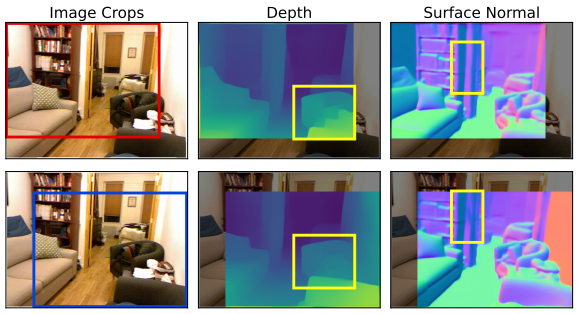


Figure 2: Current SOTA depth [18] and surface normal [12] predictors lack equivariance, which is crucial for accurate predictions. This can be seen by comparing predictions from the same scene in different crops (red and blue), where discrepancies are observed (yellow boxes).

To address this issue, I have developed an equivariant regularization technique that can be applied to any neural network [5]. This technique ensures that the model's predictions are consistent even when the input image is transformed, resulting in significant improvement in accuracy and equivariance, improved robustness to lighting changes, and better generalization to unseen environments. In ongoing work, I am exploring ways to directly incorporate illumination physics into these models to further improve robustness.

In addition to the challenge of robustness and generalization, one of the limitations of current generative models is the trade-off between image resolution and model complexity. Generating

high-resolution images often requires a large number of model parameters, resulting in increased computation time and the risk of overfitting. Moreover, these models are prone to producing artifacts that can make the generated images look fake. To address these issues, I have developed the Denoising Upsampling Residual Autoencoder (DURA), a denoising autoencoder that can learn the robust properties of images and enhance their quality for unknown noise models at test time. This approach allows for the generation of high-resolution images with minimal artifacts and can be applied to a wide range of image-denoising tasks and imaging applications [1].

To further address the challenges in generative models, I have developed techniques to improve the robustness and view generalization capabilities of generative models for novel view synthesis. This is a challenge that current methods often struggle with, as they tend to have poor predictions of appearance in unseen views during training. My approach involves using a controllable deformation model to accurately infer the underlying 3D geometry and texture of an object from a single 2D image [13]. Additionally, I have developed DIVeR [10]. This method uses deterministic estimates of the volume rendering integral to learn accurate and realistic 3D object models from small numbers of images, improving the robustness of Neural Radiance Fields (NeRF) [17]. This allows for accurate rendering of thin and translucent structures and the models produced have high rendering quality, are small in size and fast to render, and can be edited in natural ways.

My techniques for improving the robustness and generalization of AI systems can have a significant impact on practical applications, such as making self-driving cars safer and more reliable in various lighting and weather conditions and improving their ability to handle occlusions. This increases their reliability and usefulness in real-world applications.

3 Future Work

As neural rendering research [20, 11] continues to advance and large models become more prevalent, my research will play a key role in enabling the development of unified, general purpose vision models [16, 9] for comprehensive scene understanding and content creation.

Neural Rendering Engines. One of the main goals of my research is to create a neural rendering engine that uses machine learning techniques and neural networks to analyze and understand the underlying structure of 3D models, enabling it to generate high-quality images and videos. The potential benefits of using a neural rendering engine over traditional graphics engines include the ability to generate more realistic and detailed graphics, faster rendering times, and the ability to handle complex models and scenes with ease. In the future, my research will focus on combining the strengths of illumination physics and robustness to develop AI systems that can handle diverse environmental conditions and tasks with ease. In addition to developing techniques for generating diverse and realistic relighting, my research will also focus on creating targeted and controllable relighting and robust material editing methods to enable more precise manipulation of images and videos. I will also explore the generation of extreme weather data to improve the adaptability of AI systems to diverse environments. Additionally, I will work on incorporating 3D and video representations to enable more consistent view and 3D scene editing. These advancements will significantly improve the capabilities of AI systems in visual-editing applications and critical tasks such as self-driving and infrastructure assessment. My research on improving the robustness and generalization of AI systems, including the equivariant regularization technique, DURA, and DIVeR, will also be instrumental in achieving these goals.

Unified Generative-Discriminative Model. As part of my neural rendering research, I also intend to develop a unified generative-discriminative model, or "All-in-One Model," which can perform both generative tasks (such as generating new data samples or predicting the likelihood of a given sample) and discriminative tasks (such as classification or regression) effectively. These

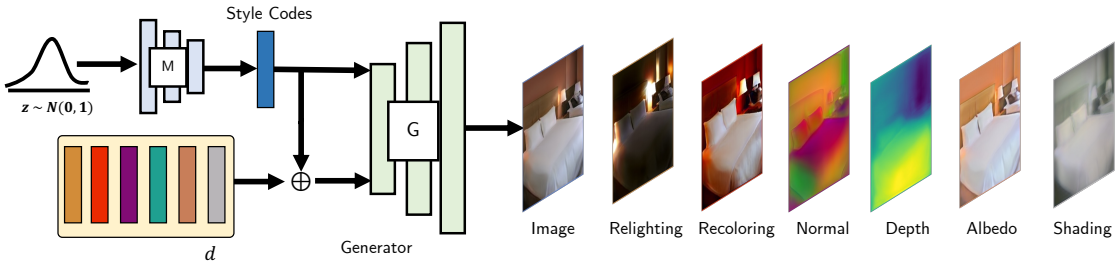


Figure 3: We demonstrate that a StyleGAN can function as a unified generative-discriminative model. Specifically, we show that it is possible to identify latent directions (d) in a pretrained StyleGAN's space which, when added to the StyleGAN's style codes, result in the generation of relighted images, recolored images, surface normal predictions, depth predictions, albedo predictions, and shading predictions, all without the need for fine-tuning or learning new weight parameters for the StyleGAN.

models have the potential to be useful in a wide range of applications because of their versatility. To objectively evaluate the capabilities of these models for scene understanding, it is important to examine their ability to capture the underlying structure of data and manipulate images based on rich, spatially, and semantically motivated representations. This is particularly important for neural rendering engines, as they need to extract important scene and semantic information. As the field of generative modeling continues to advance rapidly, my research aims to drive the development of these models and their potential applications.

In my ongoing research, I am working towards developing a unified generative-discriminative model that can perform both generative and discriminative tasks effectively. To this end, we have developed Make It So [4], a robust and near-perfect GAN inversion method that can map any given image to a StyleGAN understandable latent space. We have also found ways to extract important scene properties, such as surface normals, depth, albedo, and shading, from a StyleGAN model without the need for fine-tuning or the introduction of new learnable weight parameters (Figure 3). These advances in generative modeling have the potential to significantly improve the capabilities of AI systems in scene understanding and manipulation tasks. In the future, I will continue to research and develop techniques to improve the capabilities of unified generative-discriminative models. This will involve a range of activities, including:

1. Determining the limits of different generative model architectures in their ability to understand and comprehend the visual world.
2. Developing advanced techniques for learning disentangled representations that can more effectively capture the complex structure of real-world data and enable greater control over image generation and manipulation.
3. Investigating ways to improve the efficiency and scalability of these models, potentially through the use of techniques such as transfer learning or distillation.
4. Extending the use of these models to a variety of tasks, such as computer vision, graphics, and photography, highlighting their potential impact and versatility.
5. Developing methods for objectively assessing the performance of these models, including the use of benchmarks and metrics specifically designed for generative models.
6. Exploring the use of self-supervised learning techniques to further improve the ability of these models to learn from large amounts of unlabeled data and generalize to new environments.

In summary, my research aims to develop advanced AI systems for visual understanding and manipulation by integrating illumination physics and improving robustness and generalization. The goal of my research is to create learning systems that can handle a variety of environmental conditions by pushing the limits of what is currently possible. I envision a future where unified generative-discriminative models can revolutionize the way we view and interact with visual content, enabling the creation of a powerful neural rendering engine.

References

[1] A. Bansal, **A. Bhattad**, and D. A. Forsyth, "Dura: Denoising upsampling residual autoencoders for image enhancement," *Under Review*, 2023.

[2] **A. Bhattad**, B. Chen, S. Wang, S. R. Richter, and D. A. Forsyth, "Illuminator: Learning to correct lighting without ground truth data," *Under Review*, 2023.

[3] **A. Bhattad** and D. A. Forsyth, "Stylitgan: Enriching stylegan with illumination physics," *Under Review*, 2023.

[4] **A. Bhattad**, V. Shah, D. Hoiem, and D. A. Forsyth, "Make it so: Steering stylegan for any image inversion and editing," *Under Review*, 2023.

[5] Y. Zhong, **A. Bhattad**, Y.-X. Wang, and D. A. Forsyth, "Learning equivariant dense prediction networks," *Under Review*, 2023.

[6] **A. Bhattad** and D. A. Forsyth, "Cut-and-paste object insertion by enabling deep image prior for reshading," in *Proceedings of the International Conference on 3D Vision*, 2022.

[7] Q. A. Al-Haija, M. Gharaibeh, and A. Odeh, "Detection in adverse weather conditions for autonomous vehicles via deep learning," *AI*, vol. 3, no. 2, pp. 303–317, 2022.

[8] Z. Li *et al.*, "Physically-based editing of indoor scene lighting from a single image," *arXiv preprint arXiv:2205.09343*, 2022.

[9] J. Lu, C. Clark, R. Zellers, R. Mottaghi, and A. Kembhavi, "Unified-io: A unified model for vision, language, and multi-modal tasks," *arXiv preprint arXiv:2206.08916*, 2022.

[10] L. Wu, J. Y. Lee, **A. Bhattad**, Y.-X. Wang, and D. Forsyth, "Diver: Real-time and accurate neural radiance fields with deterministic integration for volume rendering," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 16 200–16 209.

[11] Y. Xie *et al.*, "Neural fields in visual computing and beyond," in *Computer Graphics Forum*, Wiley Online Library, vol. 41, 2022, pp. 641–676.

[12] G. Bae, I. Budvytis, and R. Cipolla, "Estimating and exploiting the aleatoric uncertainty in surface normal estimation," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 13 137–13 146.

[13] **A. Bhattad**, A. Dundar, G. Liu, A. Tao, and B. Catanzaro, "View generalization for single image textured 3d models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 6081–6090.

[14] Z. Cui, G.-J. Qi, L. Gu, S. You, Z. Zhang, and T. Harada, "Multitask aet with orthogonal tangent regularity for dark object detection," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 2553–2562.

[15] D. Forsyth and J. J. Rock, "Intrinsic image decomposition using paradigms," *IEEE transactions on pattern analysis and machine intelligence*, vol. 44, no. 11, pp. 7624–7637, 2021.

[16] T. Gupta, A. Kamath, A. Kembhavi, and D. Hoiem, "Towards general purpose vision systems," *arXiv preprint arXiv:2104.00743*, 2021.

[17] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng, "Nerf: Representing scenes as neural radiance fields for view synthesis," *Communications of the ACM*, vol. 65, no. 1, pp. 99–106, 2021.

[18] R. Ranftl, A. Bochkovskiy, and V. Koltun, "Vision transformers for dense prediction," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 12 179–12 188.

[19] T. Karras, S. Laine, M. Aittala, J. Hellsten, J. Lehtinen, and T. Aila, "Analyzing and improving the image quality of stylegan," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 8110–8119.

[20] A. Tewari *et al.*, "State of the art on neural rendering," in *Computer Graphics Forum*, Wiley Online Library, vol. 39, 2020, pp. 701–727.

[21] Y. Atoum, *Detecting objects under challenging illumination conditions*. Michigan State University, 2018.

[22] E. H. Land and J. J. McCann, "Lightness and retinex theory," *Josa*, vol. 61, no. 1, pp. 1–11, 1971.